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W4-03: Spark Actions

COMPSCI4064 (H) and COMPSCI5088 (M)

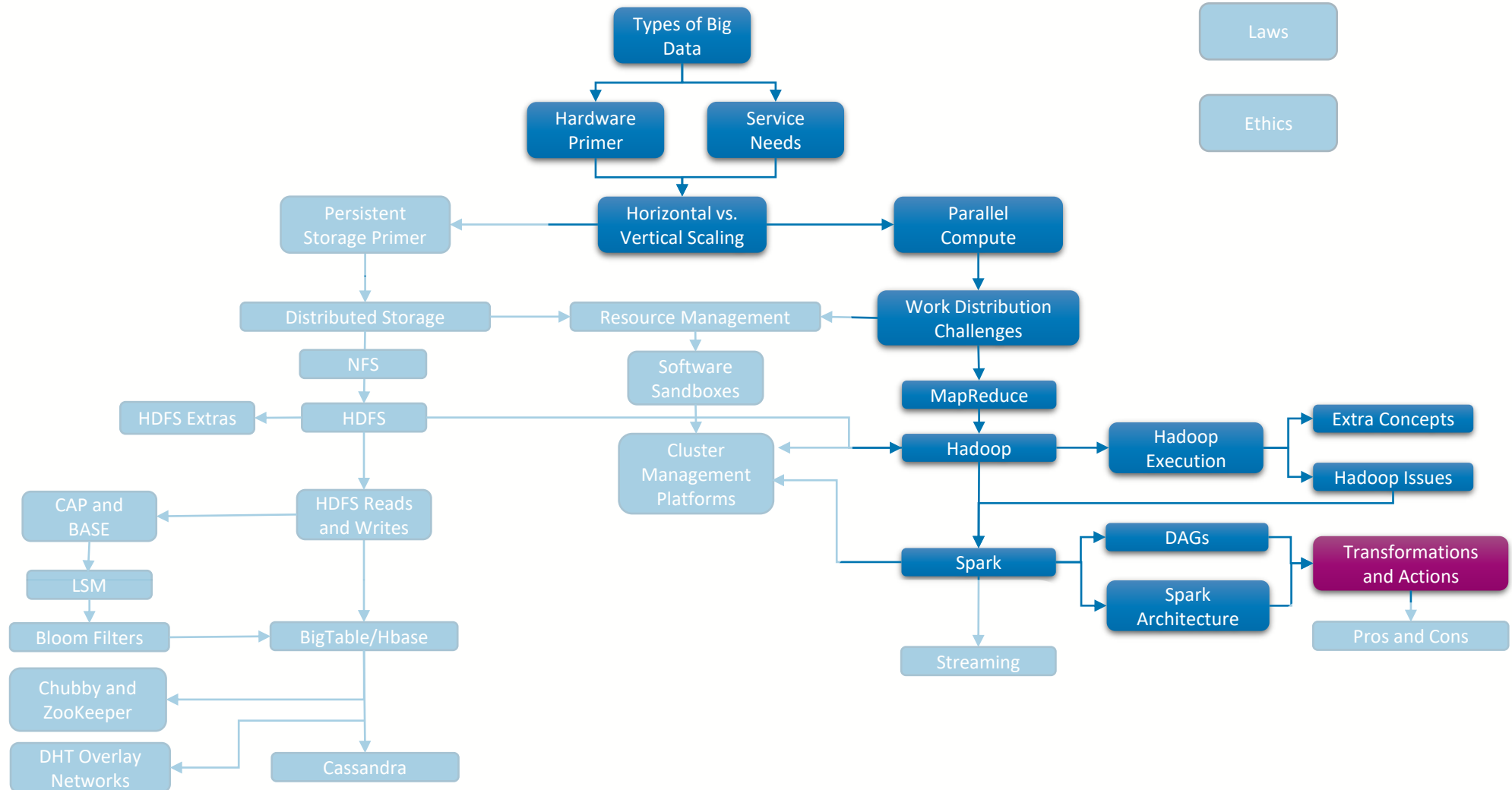
Dr. Richard McCreddie

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Where Are We?





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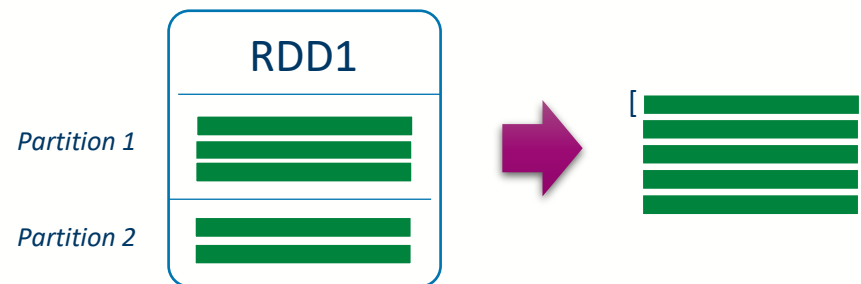
Common Actions

The ones you will use ~90% of the time



collect()

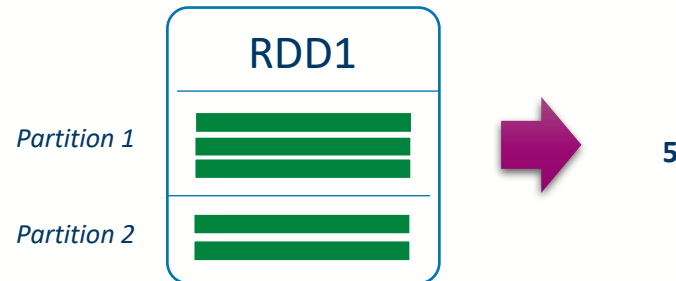
- **Description:** Returns all the elements of the dataset as an array at the driver program. This is usually useful after a filter or other operation that returns a sufficiently small subset of the data.
- **Function:** None
- **Application:** Returning data back to your machine so you can visualise or save it. Involves copying the whole RDD and sending it over the network





count()

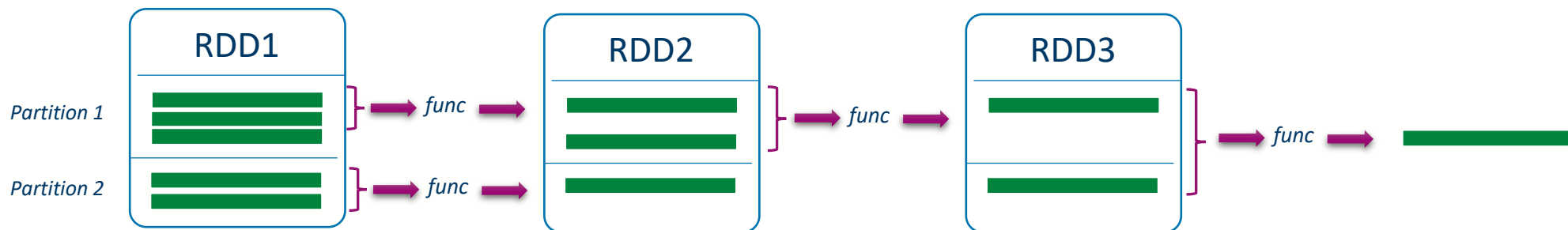
- **Description:** Return the number of elements in the dataset.
- **Function:** None
- **Application:** I usually use this when I want to trigger computation via an action, but don't want to copy all the data in the RDD to my machine.





reduce()

- **Description:** Aggregate the elements of the dataset using a function *func* (which takes two arguments and returns one). The function should be commutative and associative so that it can be computed correctly in parallel.
- **Function:** `record1, record1` $\Rightarrow$ `record1`
- **Application:** Typically used to count or aggregate information from an RDD into a single data structure, e.g. a list, map/dictionary or tree





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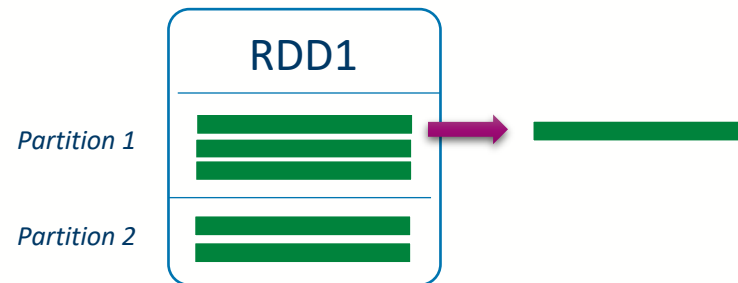
Sampling Actions

When you just want a little of the data



first()

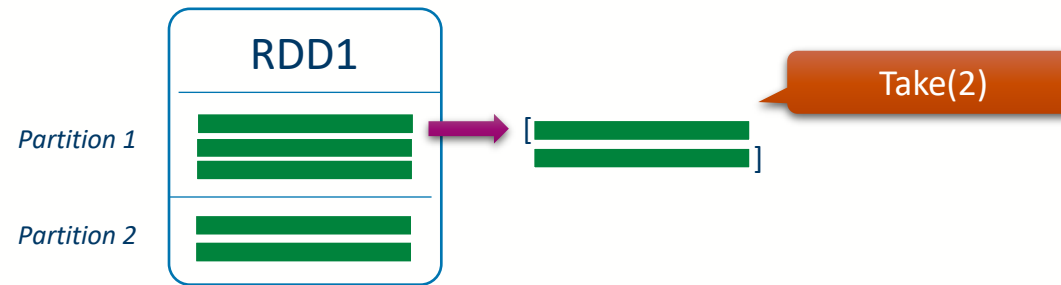
- Description: Return the first element of the dataset (similar to take(1)).
- Function: None





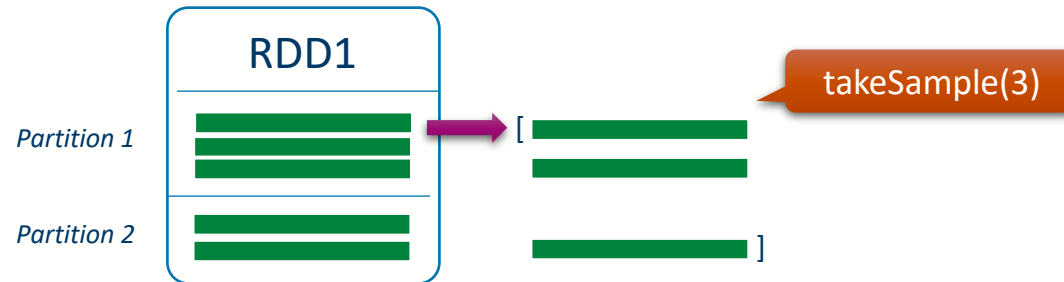
take(n)

- Description: Return an array with the first n elements of the dataset.
- Function: None



`takeSample(withReplacement, num, [seed])`

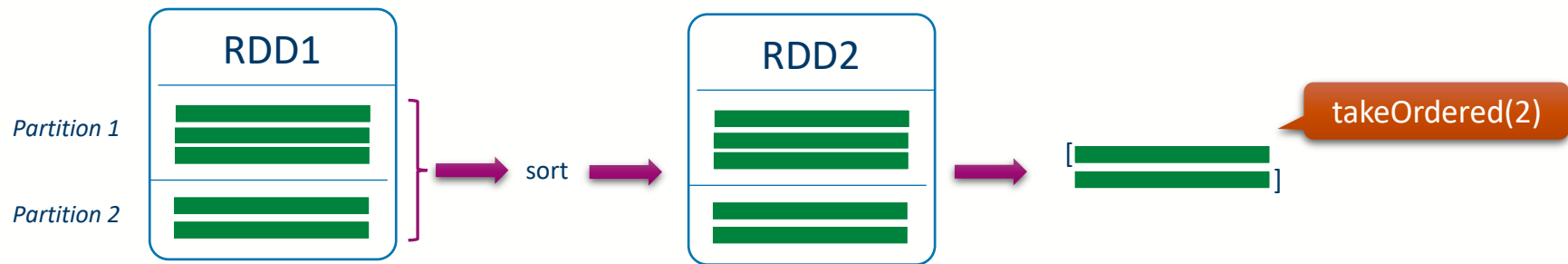
- **Description:** Return an array with a random sample of *num* elements of the dataset, with or without replacement, optionally pre-specifying a random number generator seed.
- **Function:** **None**
- **Application:** This is like `take(n)`, but is random rather than returning the first, also if you allow `withReplacement` you might get the same item multiple times





`takeOrdered(n, [ordering])`

- **Description:** Return the first n elements of the RDD using either their natural order or a custom comparator.
- **Function:** Sort
- **Application:** Again, this is like `take(n)`, but sorts the data based on their natural ordering or using a function you define, rather than returning the data in RDD order





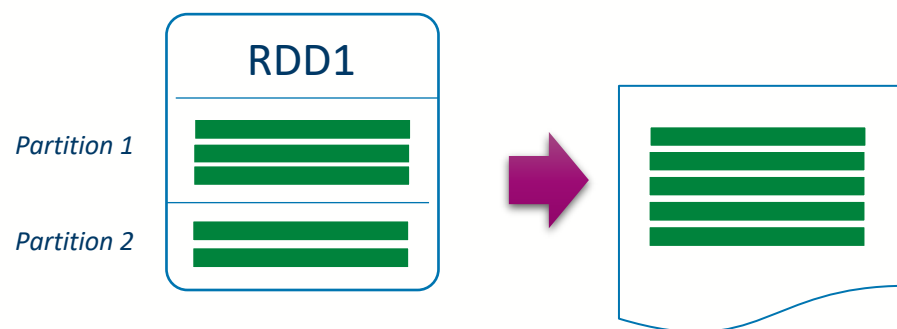
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Save Actions

When you want to keep an RDD, but its too big
for your laptop memory

`saveAsTextFile(path)`

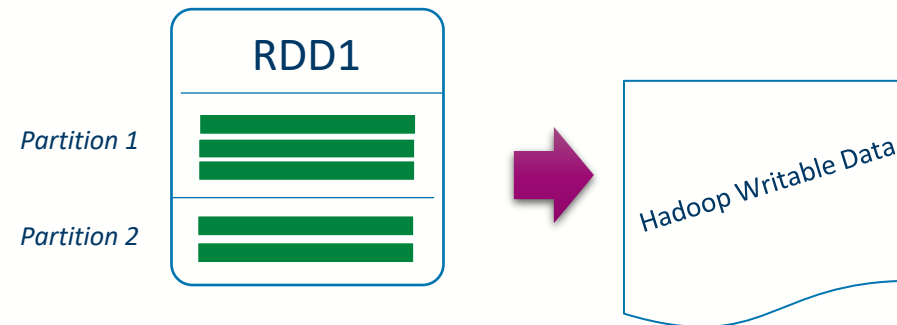
- **Description:** Write the elements of the dataset as a text file (or set of text files) in a given directory in the local filesystem, HDFS or any other Hadoop-supported file system. Spark will call `toString` on each element to convert it to a line of text in the file.
- **Function:** None





saveAsSequenceFile(*path*)

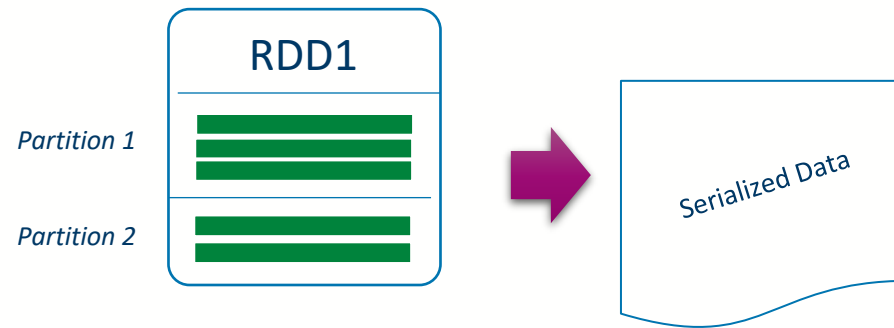
- **Description:** Write the elements of the dataset as a Hadoop SequenceFile in a given path in the local filesystem, HDFS or any other Hadoop-supported file system. This is available on RDDs of key-value pairs that implement Hadoop's Writable interface. In Scala, it is also available on types that are implicitly convertible to Writable (Spark includes conversions for basic types like Int, Double, String, etc).
- **Function:** None





`saveAsObjectFile(path)`

- **Description:** Write the elements of the dataset in a simple format using Java serialization, which can then be loaded using `SparkContext.objectFile()`.
- **Function:** None





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